Interview Questions and Answers

Q1: What is the purpose of the `np.linspace` function in NumPy?

A1: The `np.linspace` function in NumPy is used to return evenly spaced numbers over a specified interval. It generates a sequence of numbers starting from `start` to `stop` (inclusive if `endpoint=True`) with a specified number of samples (`num`).

Q2: How does the `np.random.rand` function generate random numbers?

A2: The `np.random.rand` function generates random floats in the half-open interval [0.0, 1.0). It takes dimensions as input and returns an array of the specified shape filled with random floats.

Q3: What is the difference between `np.random.randint` and `np.random.rand`?

A3: `np.random.randint` generates random integers from `low` (inclusive) to `high` (exclusive), while `np.random.rand` generates random floats in the half-open interval [0.0, 1.0). The former is used for integer values within a specified range, and the latter is used for generating random floating-point numbers.

Q4: Explain the purpose of the `np.random.seed` function and why it is important.

A4: The `np.random.seed` function sets the seed for the random number generator, ensuring that the sequence of random numbers is reproducible. By setting a seed, you can generate the same sequence of random numbers every time the code is run, which is crucial for testing and debugging.

Q5: What does the `shape` attribute of a NumPy array represent, and how can you use it?

A5: The `shape` attribute of a NumPy array returns a tuple representing the dimensions of the array. For example, if you have a 2D array with 3 rows and 4 columns, `array.shape` will return `(3, 4)`. This attribute is useful for determining the dimensions of the array and for reshaping the array using the `reshape` method.